

Floods

R1 — ISP Router

```
configure
set interfaces ethernet eth0 address 203.0.113.1/30
set system host-name R1-ISP
commit
save
```

R2 — Edge Router

```
configure
set interfaces ethernet eth0 address 203.0.113.2/30
set interfaces ethernet eth1 address 10.255.1.1/30
set system host-name R2-EDGE

set protocols static route 0.0.0.0/0 next-hop 203.0.113.1

set protocols ospf area 0 network 10.255.1.0/30

commit
save
```

R3 — Core Router

```
configure
set interfaces ethernet eth0 address 10.255.1.2/30
set interfaces ethernet eth1 address 10.255.2.1/30
set interfaces ethernet eth2 address 10.255.3.1/30
set interfaces ethernet eth3 address 10.255.4.1/30
set interfaces ethernet eth4 address 10.255.5.1/30
set interfaces ethernet eth5 address 10.0.0.2/24
set system host-name R3-CORE

set protocols ospf area 0 network 10.255.1.0/30
set protocols ospf area 0 network 10.255.2.0/30
set protocols ospf area 0 network 10.255.3.0/30
set protocols ospf area 0 network 10.255.4.0/30
set protocols ospf area 0 network 10.255.5.0/30
set protocols ospf area 0 network 10.0.0.0/24

commit
save
```

R4 — Hospital Router

```
configure
set interfaces ethernet eth0 address 10.255.2.2/30
set interfaces ethernet eth1 address 10.10.1.1/24
set system host-name R4-HOSPITAL

set protocols ospf area 0 network 10.255.2.0/30
set protocols ospf area 0 network 10.10.1.0/24

commit
save
```

R5 — Emergency Router

```
configure
set interfaces ethernet eth0 address 10.255.3.2/30
set interfaces ethernet eth1 address 10.20.1.1/24
set system host-name R5-EMERGENCY

set protocols ospf area 0 network 10.255.3.0/30
set protocols ospf area 0 network 10.20.1.0/24

commit
save
```

R6 — University Router

```
configure
set interfaces ethernet eth0 address 10.255.4.2/30
set interfaces ethernet eth1 address 10.30.1.1/24
set system host-name R6-UNIVERSITY

set protocols ospf area 0 network 10.255.4.0/30
set protocols ospf area 0 network 10.30.1.0/24

commit
save
```

R7 — Municipality Router

```
configure
set interfaces ethernet eth0 address 10.255.5.2/30
set interfaces ethernet eth1 address 10.40.1.1/24
set system host-name R7-MUNICIPAL
```

```
set protocols ospf area 0 network 10.255.5.0/30
set protocols ospf area 0 network 10.40.1.0/24

commit
save
```

Default Route toward Internet (Run on R3)

```
configure
set protocols static route 0.0.0.0/0 next-hop 10.255.1.1
commit
save
```

✓VPCS HOST COMMANDS

Hospital

```
ip 10.10.1.10/24 10.10.1.1
ip 10.10.1.11/24 10.10.1.1
ip 10.10.1.12/24 10.10.1.1
ip 10.10.1.13/24 10.10.1.1
```

Emergency

```
ip 10.20.1.10/24 10.20.1.1
ip 10.20.1.11/24 10.20.1.1
ip 10.20.1.12/24 10.20.1.1
ip 10.20.1.13/24 10.20.1.1
```

University

```
ip 10.30.1.10/24 10.30.1.1
ip 10.30.1.11/24 10.30.1.1
ip 10.30.1.12/24 10.30.1.1
ip 10.30.1.13/24 10.30.1.1
```

Municipality

```
ip 10.40.1.10/24 10.40.1.1
ip 10.40.1.11/24 10.40.1.1
ip 10.40.1.12/24 10.40.1.1
ip 10.40.1.13/24 10.40.1.1
```

Quick Verification Commands

Run anywhere:

```
show ip ospf neighbor show ip route ping 10.20.1.10 ping 8.8.8.8
```

Earthquake

pfSense (Earthquake)

Interfaces

WAN
IP: 198.51.100.2/30
GW: 198.51.100.1

LAN
IP: 10.0.0.1/24

R1 — ISP (VyOS/Cisco equivalent)

```
configure
set interfaces ethernet eth0 address 198.51.100.1/30
set system host-name R1-LTE
commit
save
```

R2 — Edge Router

```
configure
set interfaces ethernet eth0 address 198.51.100.2/30
set interfaces ethernet eth1 address 10.255.1.1/30
set interfaces ethernet eth2 address 10.255.6.1/30
set system host-name R2-EDGE

set protocols ospf area 0 network 10.255.1.0/30
set protocols ospf area 0 network 10.255.6.0/30

commit
save
```

R3 — Core Router (Mesh Hub)

```
configure
set interfaces ethernet eth0 address 10.255.1.2/30
set interfaces ethernet eth1 address 10.255.2.1/30
set interfaces ethernet eth2 address 10.255.3.1/30
set interfaces ethernet eth3 address 10.255.4.1/30
set interfaces ethernet eth4 address 10.255.5.1/30
```

```
set interfaces ethernet eth5 address 10.0.0.2/24
set system host-name R3-CORE
```

```
set protocols ospf area 0 network 10.255.1.0/30
set protocols ospf area 0 network 10.255.2.0/30
set protocols ospf area 0 network 10.255.3.0/30
set protocols ospf area 0 network 10.255.4.0/30
set protocols ospf area 0 network 10.255.5.0/30
set protocols ospf area 0 network 10.0.0.0/24
```

```
commit
save
```

R4 — Backup Router

```
configure
set interfaces ethernet eth0 address 10.255.6.2/30
set system host-name R4-BACKUP
```

```
set protocols ospf area 0 network 10.255.6.0/30
commit
save
```

R5 — Hospital

```
configure
set interfaces ethernet eth0 address 10.255.2.2/30
set interfaces ethernet eth1 address 10.10.1.1/24
set system host-name R5-HOSPITAL
```

```
set protocols ospf area 0 network 10.255.2.0/30
set protocols ospf area 0 network 10.10.1.0/24
commit
save
```

R6 — Emergency

```
configure
set interfaces ethernet eth0 address 10.255.3.2/30
set interfaces ethernet eth1 address 10.20.1.1/24
set system host-name R6-EMERGENCY
```

```
set protocols ospf area 0 network 10.255.3.0/30
set protocols ospf area 0 network 10.20.1.0/24
commit
save
```

R7 — University

```
configure
set interfaces ethernet eth0 address 10.255.4.2/30
set interfaces ethernet eth1 address 10.30.1.1/24
set system host-name R7-UNIVERSITY

set protocols ospf area 0 network 10.255.4.0/30
set protocols ospf area 0 network 10.30.1.0/24
commit
save
```

R8 — Office

```
configure
set interfaces ethernet eth0 address 10.255.5.2/30
set interfaces ethernet eth1 address 10.40.1.1/24
set system host-name R8-OFFICE

set protocols ospf area 0 network 10.255.5.0/30
set protocols ospf area 0 network 10.40.1.0/24
commit
save
```

VPCS Commands

Hospital

```
ip 10.10.1.10/24 10.10.1.1
ip 10.10.1.11/24 10.10.1.1
ip 10.10.1.12/24 10.10.1.1
ip 10.10.1.13/24 10.10.1.1
```

Emergency

```
ip 10.20.1.10/24 10.20.1.1
ip 10.20.1.11/24 10.20.1.1
ip 10.20.1.12/24 10.20.1.1
ip 10.20.1.13/24 10.20.1.1
```

University

```
ip 10.30.1.10/24 10.30.1.1
ip 10.30.1.11/24 10.30.1.1
ip 10.30.1.12/24 10.30.1.1
```

```
ip 10.30.1.13/24 10.30.1.1
```

Office

```
ip 10.40.1.10/24 10.40.1.1  
ip 10.40.1.11/24 10.40.1.1  
ip 10.40.1.12/24 10.40.1.1  
ip 10.40.1.13/24 10.40.1.1
```

Internet Route (Run on Core)

```
configure set protocols static route 0.0.0.0/0 next-hop 10.255.1.1 commit  
save
```